Rongheng Qi

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EDUCATION BACKGROUND

University College London — London, United Kingdom
Master of Research in Risk and Disaster Reduction — Sept. 2025 - Dec. 2026 (Expected)
Grade: 70.70/100 (Distinction)

China University of Geosciences — Wuhan, China
Bachelor of Management Science in Emergency Management — Sept. 2021 - Jun. 2025
Grade: 90.31/100 (Rank: 1/28)

RESEARCH INTERESTS

- Disaster Risk Reduction
- Emergency Response
- Machine Learning
- Geospatial Analysis
- Geographic Information System (GIS)
- Geospatial Artificial Intelligence (GeoAI)

PATENTS

[1] Xie, Z., Xu, J., & **Qi, R.** (2024). Multifunctional Outdoor Tent. China Utility Model Patent, CN 221990046U. https://patents.google.com/patent/CN221990046U/en

[2] Xu, J., Xie, Z., & **Qi, R.** (2024). Anti-toppling Base and Road Warning Post. China Utility Model Patent, CN 221919100U. https://patents.google.com/patent/CN221919100U/en

RESEARCH EXPERIENCE

University College London — London, United Kingdom
Master's Dissertation | Adviser: *Prof. Saman Ghaffarian* — Apr. 2026 - Present

- Develop a building-scale flood risk assessment framework integrating urban flood susceptibility, building exposure, and building vulnerability.
- Integrate multi-source remote sensing and urban data for flood susceptibility modeling, with deep learning applied to building function identification and vulnerability assessment.
- Conduct spatial analysis and assess risk patterns across more than 120,000 buildings to support targeted flood risk management and emergency planning.

University College London — London, United Kingdom
Research Assistant | Adviser: *Prof. Saman Ghaffarian* — Mar. 2026 - Present

- Develop a post-disaster building damage classification framework by adapting DeepLabV3+ with a MobileNetV3 backbone on the xBD dataset.
- Apply XAI methods including Grad-CAM, HiRes-CAM, Integrated Gradients, and Attention-Aware Layer-Wise Relevance Propagation to interpret model predictions and examine the visual evidence driving classifications.
- Compare the performance of four XAI approaches across faithfulness, localization, stability, and computational efficiency.

China University of Geosciences — Wuhan, China

Bachelor's Dissertation | Adviser: *Prof. Banxiao Ruan* Jun. 2024 - Jun. 2025

- Develop an urban pluvial flood simulation framework integrating stochastic rainfall modeling and hydrological methods to estimate road inundation under extreme rainfall scenarios.
- Apply the Probabilistically Enhanced Two-Step Floating Catchment Area model to assess 15-minute emergency medical accessibility between schools and hospitals in Wuhan under flood disruption.
- Analyze mismatches between emergency supply and demand, revealing growing inequality and accessibility failures to inform equitable resource allocation and emergency response planning.

China University of Geosciences Wuhan, China

Research Assistant | Adviser: *Prof. Banxiao Ruan* & *Prof. Shixiang Li* May. 2023 - Jan. 2025

- Develop an urban resilience evaluation system for 110 cities using socio-ecological indicators and time-varying entropy weighting to capture spatiotemporal dynamics over a decade.
- Analyze regional disparities, spatial clustering, and evolutionary patterns in urban resilience using inequality decomposition and spatial autocorrelation analysis.
- Present research findings at the *First International Conference on Meteorological Linked Disasters and Emergency Management*.

Peking University Shenzhen Graduate School Remote

Research Assistant | Adviser: *Prof. Pengjun Zhao* & *Prof. Qiyang Liu* Jul. 2022 - Sept. 2022

- Perform a structured analysis of work resumption and production recovery policies across seven major cities in China, focusing on economic, transport, and public health measures.
- Clean, organize, and classify policy documents totaling more than 100,000 words to support post-pandemic policy evaluation.

WORK EXPERIENCE

PIESAT Information Technology Co., Ltd Wuhan, China

Disaster Analyst Intern | Emergency Management Department Dec. 2024 - Feb. 2025

- Contribute to China's nationwide disaster risk survey by conducting multi-hazard risk assessments for four Hubei cities and producing 20+ standardized zoning maps on hazard intensity, exposure, and vulnerability.
- Write analytical reports on disaster prevention and mitigation strategies, contributing to 8 official deliverables used in provincial disaster governance and planning.

Natural Resources and Planning Bureau of Badong County Enshi, China

Geospatial Analyst Intern | Disaster Prevention Department Jul. 2023 - Aug. 2023

- Support the Hubei Hydrogeology Engineering Geology Brigade in field investigations of active landslides, operating UAVs and GPS devices to collect geospatial and terrain data for hazard assessment.
- Process Sentinel-2 and Landsat-8 imagery to extract vegetation indices, surface water distribution, and terrain characteristics, contributing to hazard diagnostics.

SKILLS

Computer: Python; ArcGIS Pro; Google Earth Engine; RStudio; SPSS; Overleaf

Languages: Mandarin (native), English (fluent, IELTS: 7.0, L:7.0/ R:7.0/ W:6.5/ S:6.5)